1.46 2D Vectors

Worked Solutions

1 a)

$3(2\mathbf{i}+\mathbf{j}) = 6\mathbf{i}+3\mathbf{j}$.

1 b)

$2(3\mathbf{j}) + (-\mathbf{i}-4\mathbf{j}) = -\mathbf{i}+2\mathbf{j}$.

1 c)

$2\mathbf{i}+\mathbf{j}-2(-\mathbf{i}-4\mathbf{j}) = 4\mathbf{i}+9\mathbf{j}$.

2 a)

$\overrightarrow{AB} = \overrightarrow{AO}+\overrightarrow{OB} = -\mathbf{a}+\mathbf{b}$.

2 b)

$\overrightarrow{CB} = \overrightarrow{CO}+\overrightarrow{OB} = -\mathbf{c}+\mathbf{b}$.

2 c)

$\overrightarrow{AC} = \overrightarrow{AO}+\overrightarrow{OC} = -\mathbf{a}+\mathbf{c}$.

3 a)

$\overrightarrow{OP} = \frac{1}{2}\mathbf{a}$.

3 b)

$\overrightarrow{OQ} = \frac{1}{2}\mathbf{b}$.

3 c)

$\overrightarrow{OR} = \mathbf{b}+\frac{1}{2}\mathbf{a}$.

3 d)

$\overrightarrow{OS} = \mathbf{a}+\frac{1}{2}\mathbf{b}$.

3 e)

$\overrightarrow{BQ} = -\frac{1}{2}\mathbf{b}$.

3 f)

$\overrightarrow{SR} = \frac{1}{2}\mathbf{b}-\frac{1}{2}\mathbf{a}$.

3 g)

$\overrightarrow{RP} = -\frac{1}{2}\mathbf{a}-\mathbf{b}+\frac{1}{2}\mathbf{a} = -\mathbf{b}$.

3 h)

$\overrightarrow{AQ} = -\mathbf{a} + \frac{1}{2}\mathbf{b}$.

4 a)

$4\mathbf{a} = -2(-2\mathbf{a}) \Rightarrow$ Parallel.

4 b)

$2\mathbf{a} + \mathbf{b} = 2(\mathbf{a} + \frac{1}{2}\mathbf{b}) \Rightarrow$ Parallel.

4 c)

$3\mathbf{a} - \mathbf{b} \neq -1(-3\mathbf{a} - \mathbf{b}) \Rightarrow$ Not parallel.

4 d)

$-2\mathbf{a} - 4\mathbf{b} = -\frac{1}{4}(8\mathbf{a} + 24\mathbf{b}) \Rightarrow$ Not parallel.

5 a)

$|\mathbf{a}| = \sqrt{4^2 + 5^2} = \sqrt{41}$.

5 b)

$|\mathbf{b}| = \sqrt{7^2 + 1^2} = 5\sqrt{2}$.

5 c)

$|\mathbf{c}| = \sqrt{3^2 + 1^2} = \sqrt{10}$.

5 d)

$|\mathbf{d}| = \sqrt{p^2 + q^2}$.

6 a)

$|\mathbf{a}| = \sqrt{2^2 + 12^2} = 2\sqrt{37}$.

$\tan^{-1}(\frac{12}{2}) = 80.5°$.

$90° - \tan^{-1}(\frac{12}{2})° = 9°$.

$\therefore 2\sqrt{37}$ on bearing 009°.

6 b)

$|\mathbf{b}| = \sqrt{5^2 + 3^2} = \sqrt{34}$.

$\tan^{-1}(\frac{3}{5}) = 30.96375653\ldots°$.

$270° + \tan^{-1}(\frac{3}{5}) = 301°$.

$\therefore \sqrt{34}$ on bearing 301°.

6 c)

$|\mathbf{c}| = \sqrt{10^2 + 5^2} = 5\sqrt{5}$.

$\tan^{-1}(\frac{5}{10}) = 26.56505118\ldots°$.

$270° - \tan^{-1}(\frac{5}{10}) = 243°$.

$\therefore 5\sqrt{5}$ on bearing 243°.

6 d)

$|\mathbf{d}| = \sqrt{8^2 + 15^2} = 17.$

$\tan^{-1}(\frac{15}{8}) = 61.92751306\ldots^\circ.$

$90^\circ + \tan^{-1}(\frac{15}{8}) = 152^\circ.$

$\therefore 17$ on bearing $152^\circ.$

7 a)

$p = 12\cos 40 = 9.19$

$q = 12\sin 40 = 7.71$

$9.19\mathbf{i} + 7.71\mathbf{j}.$

7 b)

$5\cos 24 = 4.57$

$5\sin 24 = 2.03$

$p = -4.57,\ q = 2.03$

$-4.57\mathbf{i} + 2.03\mathbf{j}.$

7 c)

$180 - 124 = 56$

$18\cos 56 = 10.1$

$18\sin 56 = 14.9$

$p = -10.1,\ q = -14.9$

$-10.1\mathbf{i} - 14.9\mathbf{j}.$

7 d)

$360 - 290 = 70$

$55\cos 70 = 18.8$

$55\sin 70 = 51.7$

$p = 18.8,\ q = -51.7$

$18.8\mathbf{i} - 51.7\mathbf{j}.$

8 a)

$|\mathbf{a}| = \sqrt{2^2 + 3^2} = \sqrt{13}.$

$\frac{1}{\sqrt{13}}(2\mathbf{i} + 3\mathbf{j}).$

8 b)

$|\mathbf{b}| = \sqrt{1^2 + 4^2} = \sqrt{17}.$

$\frac{1}{\sqrt{17}}\begin{pmatrix} -1 \\ 4 \end{pmatrix}.$

8 c)

$|\mathbf{c}| = \sqrt{5^2 + 8^2} = \sqrt{89}$.

$\frac{1}{\sqrt{89}}(5\mathbf{i} - 8\mathbf{j})$.

9 a)

$|\mathbf{p}| = \sqrt{2^2 + 6^2} = 2\sqrt{10}$.

$\mathbf{a} = \frac{1}{2\sqrt{10}}(-2\mathbf{i} + 6\mathbf{j}) \times 3\sqrt{10} = -3\mathbf{i} + 9\mathbf{j}$.

9 b)

$|\mathbf{p}| = \sqrt{2^2 + 7^2} = \sqrt{53}$.

$\mathbf{a} = \frac{1}{\sqrt{53}}(2\mathbf{i} + 7\mathbf{j}) \times \frac{1}{2}\sqrt{53} = \mathbf{i} + \frac{7}{2}\mathbf{j}$.

9 c)

$|\mathbf{p}| = \sqrt{1^2 + 8^2} = \sqrt{65}$.

$\mathbf{a} = \frac{1}{\sqrt{65}}(\mathbf{i} - 8\mathbf{j}) \times 3\sqrt{65} = 3\mathbf{i} - 24\mathbf{j}$.

10 a)

$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

10 b)

$\begin{pmatrix} -1 \\ 6 \end{pmatrix}$.

10 c)

$\begin{pmatrix} p \\ q \end{pmatrix}$.

11 a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$.

11 b)

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} 3 \\ -10 \end{pmatrix} - \begin{pmatrix} -2 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ -16 \end{pmatrix}$.

11 c)

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -10 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -12 \end{pmatrix}$.

11 d)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix}1\\2\end{pmatrix} - \begin{pmatrix}-2\\6\end{pmatrix} = \begin{pmatrix}3\\-4\end{pmatrix}$.

12 a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}-5\\2\end{pmatrix} - \begin{pmatrix}3\\-1\end{pmatrix} = \begin{pmatrix}-8\\3\end{pmatrix}$.

$|\overrightarrow{AB}| = \sqrt{8^2 + 3^2} = \sqrt{73}$.

12 b)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 9\mathbf{i} + 4\mathbf{j} - (2\mathbf{i} - 7\mathbf{j}) = 7\mathbf{i} + 11\mathbf{j}$.

$|\overrightarrow{AB}| = \sqrt{7^2 + 11^2} = \sqrt{170}$.

13 a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}3\\6\end{pmatrix} - \begin{pmatrix}1\\-5\end{pmatrix} = \begin{pmatrix}2\\11\end{pmatrix}$.

$|\overrightarrow{AB}| = \sqrt{2^2 + 11^2} = 5\sqrt{5}$.

$\frac{1}{5\sqrt{5}} \begin{pmatrix}2\\11\end{pmatrix}$.

13 b)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = -3\mathbf{i} + 5\mathbf{j} - (5\mathbf{i} + 2\mathbf{j}) = -8\mathbf{i} + 3\mathbf{j}$.

$|\overrightarrow{AB}| = \sqrt{8^2 + 3^2} = \sqrt{73}$.

$\frac{1}{\sqrt{73}}(-8\mathbf{i} + 3\mathbf{j})$.

14 a)

$90°$.

14 b)

$\tan^{-1}(\frac{7}{2}) = 74.0546041\ldots$

$\tan^{-1}(\frac{3}{3}) = 45$

$74.0546041\ldots - 45 = 29.1°$.

14 c)

$\tan^{-1}(\frac{10}{2}) = 78.69006753\ldots$

$\tan^{-1}(\frac{3}{8}) = 20.55604522\ldots$

$78.69006753\ldots - 20.55604522\ldots = 58.1°$.

14 d)

$\tan^{-1}(\frac{5}{3}) = 59.03624347\ldots$

$\Rightarrow 90 + 59.03624347\ldots = 149.0°$.

14 e)

$\tan^{-1}(\frac{2}{1}) = 63.43494882\ldots$

$\tan^{-1}(\frac{7}{1}) = 81.86989765\ldots$

$\Rightarrow 180 - 81.869\ldots = 98.13010235\ldots$

$98.1301\ldots + 63.43494882\ldots = 161.6°.$

15)

$\sqrt{1^2+2^2} = \sqrt{5}$ and $\sqrt{3^2+1^2} = \sqrt{10}$.

$\overrightarrow{AC} = \begin{pmatrix}1\\-2\end{pmatrix} + \begin{pmatrix}3\\-1\end{pmatrix} = \begin{pmatrix}4\\-3\end{pmatrix}.$

$\sqrt{4^2+3^2} = 5.$

$5^2 = (\sqrt{5})^2 + (\sqrt{10})^2 - 2\sqrt{5}\sqrt{10}\cos\theta.$

$25 = 5 + 10 - 10\sqrt{2}\cos\theta.$

$10 = -10\sqrt{2}\cos\theta.$

$\cos\theta = -\dfrac{10}{10\sqrt{2}} = -\dfrac{1}{\sqrt{2}} = -\dfrac{\sqrt{2}}{2}$ as req.

16)

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix}-5\\1\end{pmatrix} - \begin{pmatrix}4\\8\end{pmatrix} = \begin{pmatrix}-9\\-7\end{pmatrix}.$

$\overrightarrow{AD} = \begin{pmatrix}-9\\-7\end{pmatrix}.$

$\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD} = \begin{pmatrix}3\\2\end{pmatrix} + \begin{pmatrix}-9\\-7\end{pmatrix} = \begin{pmatrix}-6\\-5\end{pmatrix}.$

17 a)

$\overrightarrow{AB} = \begin{pmatrix}0\\7\end{pmatrix} - \begin{pmatrix}-8\\2\end{pmatrix} = \begin{pmatrix}8\\5\end{pmatrix}.$

$\overrightarrow{BC} = \begin{pmatrix}8\\12\end{pmatrix} - \begin{pmatrix}0\\7\end{pmatrix} = \begin{pmatrix}8\\5\end{pmatrix}.$

$\therefore$ collinear.

17 b)

$\overrightarrow{AB} = \begin{pmatrix}4\\-5\end{pmatrix} - \begin{pmatrix}-5\\4\end{pmatrix} = \begin{pmatrix}9\\-9\end{pmatrix}.$

$\overrightarrow{BC} = \begin{pmatrix}7\\-8\end{pmatrix} - \begin{pmatrix}4\\-5\end{pmatrix} = \begin{pmatrix}3\\-3\end{pmatrix}.$

$\overrightarrow{AB} = 3\overrightarrow{BC}$ $\therefore$ collinear.

17 c)

$\overrightarrow{AB} = \begin{pmatrix}-3\\13\end{pmatrix} - \begin{pmatrix}-10\\17\end{pmatrix} = \begin{pmatrix}7\\-4\end{pmatrix}.$

$\overrightarrow{BC} = \begin{pmatrix}5\\8\end{pmatrix} - \begin{pmatrix}-3\\13\end{pmatrix} = \begin{pmatrix}8\\-5\end{pmatrix}$.

$\therefore$ not collinear.

18)

$\overrightarrow{OA} = -5\mathbf{i} + 13\mathbf{j}$, $\overrightarrow{OM} = -\mathbf{i} + 4\mathbf{j}$.

$\overrightarrow{AM} = \overrightarrow{OM} - \overrightarrow{OA} = -\mathbf{i} + 4\mathbf{j} - (-5\mathbf{i} + 13\mathbf{j}) = 4\mathbf{i} - 9\mathbf{j}$.

$\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AM} = -5\mathbf{i} + 13\mathbf{j} + 2(4\mathbf{i} - 9\mathbf{j})$

$= 3\mathbf{i} - 5\mathbf{j}$.

19 a)

$\overrightarrow{AB} = \begin{pmatrix}4\\3\end{pmatrix} - \begin{pmatrix}-3\\7\end{pmatrix} = \begin{pmatrix}7\\-4\end{pmatrix}$.

$\overrightarrow{CD} = \begin{pmatrix}p\\8\end{pmatrix} - \begin{pmatrix}1\\k\end{pmatrix} = \begin{pmatrix}7\\-4\end{pmatrix}$.

$p = 8$, $k = 12$.

19 b)

AB is parallel to CD and has same length.

$\overrightarrow{AC} = \begin{pmatrix}1\\12\end{pmatrix} - \begin{pmatrix}-3\\7\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix}$.

$\overrightarrow{BD} = \begin{pmatrix}8\\8\end{pmatrix} - \begin{pmatrix}4\\3\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix}$.

AC is parallel to BD and has same length but different to AB.

Therefore, it is a parallelogram.

20 a i)

$\overrightarrow{OC} = \overrightarrow{AB} = \mathbf{b} - \mathbf{a}$.

20 a ii)

Midpoint of $\overrightarrow{AB}$:

$\overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} = \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}\mathbf{b} + \frac{1}{2}\mathbf{a}$.

20 a iii)

Midpoint of $\overrightarrow{OC}$:

$\frac{1}{2}\overrightarrow{OC} = \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a}$.

20 b)

The centre is where the two diagonals intersect:

$\frac{1}{2}\overrightarrow{OB} = \frac{1}{2}\mathbf{b}$.

$\frac{1}{2}\overrightarrow{AC} = \frac{1}{2}\overrightarrow{OC} - \frac{1}{2}\overrightarrow{OA} = \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a} - \frac{1}{2}\mathbf{a} = \frac{1}{2}\mathbf{b}$.

And $\frac{1}{2}\overrightarrow{PQ} = \frac{1}{2}\overrightarrow{OC} + \frac{1}{2}\overrightarrow{OA} = \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{a} = \frac{1}{2}\mathbf{b}$.

Therefore, the midpoint of $\overrightarrow{PQ}$ is at the centre of the parallelogram.

21 a)

$\overrightarrow{AB} = \overrightarrow{DC}$.

$4\mathbf{i} + 5\mathbf{j} = 5\mathbf{i} + \mathbf{j} - (k\mathbf{i} - 4\mathbf{j}) \Rightarrow k = 1$.

21 b)

$5\mathbf{i} + \mathbf{j} - 5\mathbf{i} + 4\mathbf{j} = 5\mathbf{j}$.

$\mathbf{i} - 4\mathbf{j} - 5\mathbf{i} + 4\mathbf{j} = -4\mathbf{i}$.

Coordinates are:

$A(-4, 0)$ and $B(0, 5)$.

22 a)

$\overrightarrow{AB} = \begin{pmatrix}1\\6\end{pmatrix} - \begin{pmatrix}5\\2\end{pmatrix} = \begin{pmatrix}-4\\4\end{pmatrix} \Rightarrow \frac{1}{2}\overrightarrow{AB} = \begin{pmatrix}-2\\2\end{pmatrix}$.

$\overrightarrow{OP} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} = \begin{pmatrix}5\\2\end{pmatrix} + \begin{pmatrix}-2\\2\end{pmatrix} = \begin{pmatrix}3\\4\end{pmatrix}$.

$P(3, 4)$.

$\overrightarrow{BC} = \begin{pmatrix}-5\\4\end{pmatrix} - \begin{pmatrix}1\\6\end{pmatrix} = \begin{pmatrix}-6\\-2\end{pmatrix} \Rightarrow \frac{1}{2}\overrightarrow{BC} = \begin{pmatrix}-3\\-1\end{pmatrix}$.

$\overrightarrow{OQ} = \overrightarrow{OB} + \frac{1}{2}\overrightarrow{BC} = \begin{pmatrix}1\\6\end{pmatrix} + \begin{pmatrix}-3\\-1\end{pmatrix} = \begin{pmatrix}-2\\5\end{pmatrix}$.

$Q(-2, 5)$.

22 b)

$\overrightarrow{PQ} = \begin{pmatrix}-2\\5\end{pmatrix} - \begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}-5\\1\end{pmatrix}$.

$\overrightarrow{AC} = \begin{pmatrix}-5\\4\end{pmatrix} - \begin{pmatrix}5\\2\end{pmatrix} = \begin{pmatrix}-10\\2\end{pmatrix}$.

$\therefore \overrightarrow{PQ} = \frac{1}{2}\overrightarrow{AC}$ as req.

22 c)

$\overrightarrow{DB} = \begin{pmatrix}1\\6\end{pmatrix} - \begin{pmatrix}d_x\\d_y\end{pmatrix} = \begin{pmatrix}1-d_x\\6-d_y\end{pmatrix}$.

$\sqrt{(1-d_x)^2 + (6-d_y)^2} = \sqrt{74}$.

$(1-d_x)^2 + (6-d_y)^2 = 74$.

$\overrightarrow{AD} = \begin{pmatrix}d_x\\d_y\end{pmatrix} - \begin{pmatrix}5\\2\end{pmatrix} = \begin{pmatrix}d_x-5\\d_y-2\end{pmatrix}$ is parallel to $\begin{pmatrix}-6\\-2\end{pmatrix}$.

$d_x - 5 = -6k$ and $d_y - 2 = -2k$.

$(1 - 5 + 6k)^2 + (6 - 2 + 2k)^2 = 74$.

$(-4 + 6k)^2 + (4 + 2k)^2 = 74$.

$16 - 48k + 36k^2 + 16 + 16k + 4k^2 = 74.$

$40k^2 - 32k - 42 = 0.$

$k = \frac{3}{2}$ or $k = -\frac{7}{10}$.

$d_x - 5 = -6 \times \frac{3}{2}$, $d_y - 2 = -2 \times \frac{3}{2} \Rightarrow (-4, -1)$.

$d_x - 5 = -6 \times -\frac{7}{10}$, $d_y - 2 = -2 \times -\frac{7}{10} \Rightarrow (\frac{46}{5}, \frac{17}{5})$.

23)

$\overrightarrow{AB} = \begin{pmatrix} 2 \\ 10 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 8 \end{pmatrix}$.

Let M be the midpoint of AB:

$\overrightarrow{OM} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 4 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \\ 6 \end{pmatrix}$.

Bisector of triangle is $\begin{pmatrix} 1.5 \\ 6 \end{pmatrix} + k \begin{pmatrix} 8 \\ -1 \end{pmatrix}$.

$\overrightarrow{OC} = \begin{pmatrix} \frac{3}{2} + 8k \\ 6 - k \end{pmatrix}$.

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} \frac{3}{2} + 8k - 1 \\ 6 - k - 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} + 8k \\ 4 - k \end{pmatrix}$.

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} \frac{3}{2} + 8k - 2 \\ 6 - k - 10 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} + 8k \\ -4 - k \end{pmatrix}$.

$\frac{1}{2} + 8k = -(-4 - k) \Rightarrow \frac{1}{2} + 8k = 4 + k \Rightarrow k = \frac{1}{2}$

which gives $(5.5, 5.5)$.

$-\frac{1}{2} + 8k = -(4 - k) \Rightarrow -\frac{1}{2} + 8k = -4 + k \Rightarrow k = -\frac{1}{2}$ which gives $(-2.5, 6.5)$.